# Lab1 Spice Simulation of Setup Time Estimation

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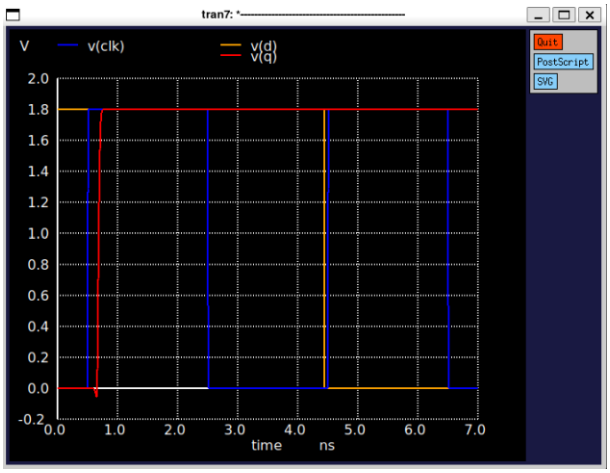
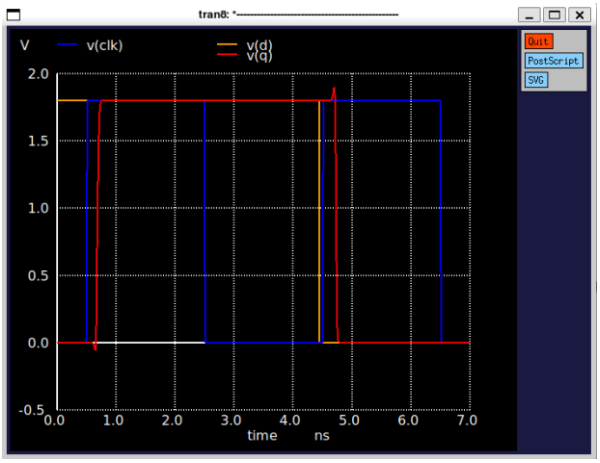
| -- | Voltage (V) | $T_{pcq}$ (ps) | $T_{setup}$ (ps) |
|---|---|---|---|
| tt | 1.8 | 225.65 | 64 |
| tt | 1.98 | 174.97 | 52 |
| tt | 1.62 | 437.2 | 84 |
| ff | 1.8 | 153.4 | 44 |
| ff | 1.98 | 128.2 | 38 |
| ff | 1.62 | 195.3 | 54 |
| ss | 1.8 | 553.9 | 108 |
| ss | 1.98 | 295.2 | 82 |
| ss | 1.62 | 689.5 | 158 |
| fs | 1.8 | 248.2 | 44 |
| fs | 1.98 | 191.7 | 38 |
| fs | 1.62 | 316.9 | 54 |
| sf | 1.8 | 317.7 | 102 |
| sf | 1.98 | 187.8 | 78 |
| sf | 1.62 | 450.5 | 144 |

Please paste the waveform and the screenshot of the measurement result

(15 sets of screenshot)

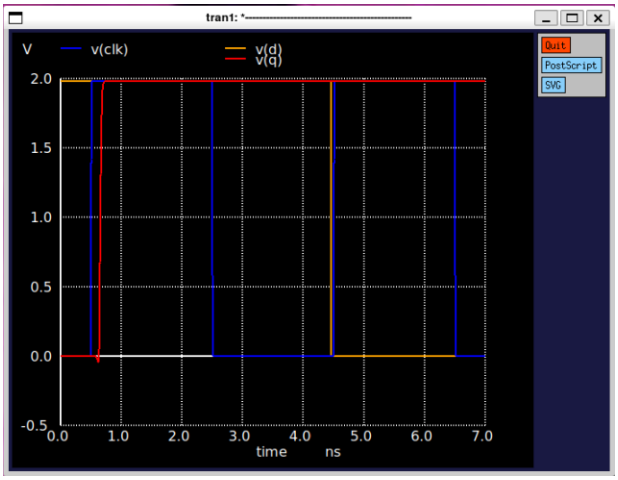
1. tt corner & 1.8V (please remove the example screanshots)

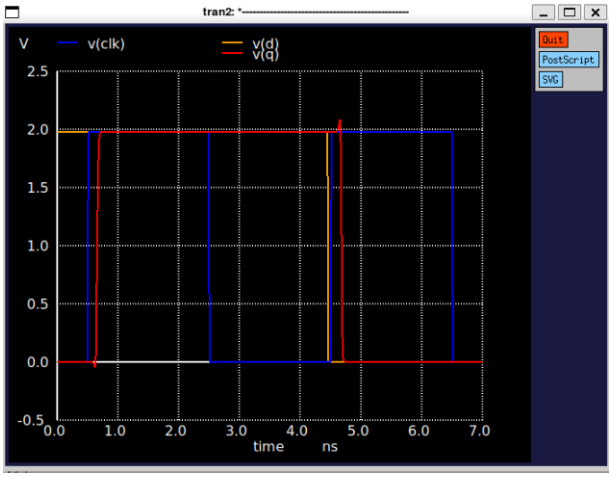



```
 Reference value :  6.98255e-09
No. of Data Rows : 70038
tdc_delay = 6.200000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.000000e-01 rise=2 targ v(q) val=9.000000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                        1.8
xdff.q_n                                   1.8
d                                          1.8
q                                    5.78959e-09
clk                                          0
y                                          1.8
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -5.59378e-11

 Reference value :  6.95365e-09
No. of Data Rows : 70038
tdc_delay = 6.400000e-11
tpcq                 =  2.256499e-10 targ=  4.730650e-09 trig=  4.505000e-09
Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------
```

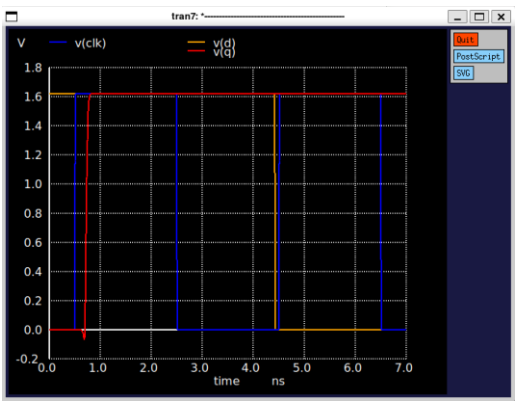
2. tt corner & 1.98V

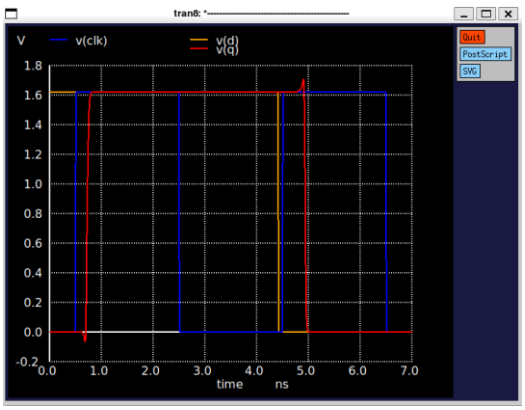



```
clk                                          0
y                                         1.98
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -8.27617e-11

 Reference value :  6.95515e-09
No. of Data Rows : 70038
Warning: Missing charsets in String to FontSet conversion
tdc_delay = 5.000000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.900000e-01 rise=2 targ v(q) val=9.900000e-01 fall=1 failed!

Reset re-loads circuit *----------------------------------------------

Circuit: *----------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.98
xdff.q_n                                  1.98
d                                         1.98
q                                    7.8052e-09
clk                                          0
y                                         1.98
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -8.27617e-11

 Reference value :  6.97585e-09
No. of Data Rows : 70038
tdc_delay = 5.200000e-11
tpcq                =  1.749742e-10 targ=  4.679974e-09 trig=  4.505000e-09
Reset re-loads circuit *----------------------------------------------
```

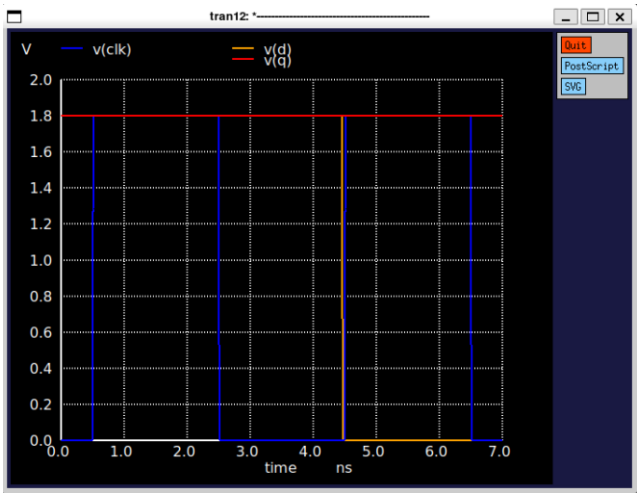
## 3. tt corner & 1.62V

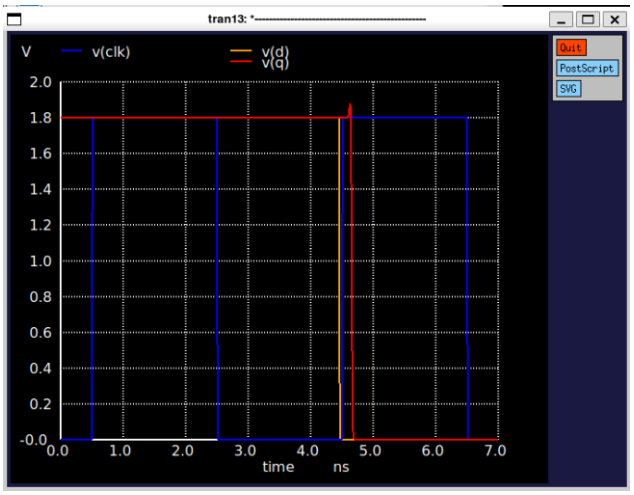


```
 Reference value :  6.94805e-09
No. of Data Rows : 70038
tdc_delay = 8.200000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=8.100000e-01 rise=2 targ v(q) val=8.100000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.62
xdff.q_n                                  1.62
d                                         1.62
q                                    4.90854e-09
clk                                          0
y                                         1.62
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -4.28342e-11

 Reference value :  6.96845e-09
No. of Data Rows : 70038
tdc_delay = 8.400000e-11
tpcq                =  4.372279e-10 targ=  4.942228e-09 trig=  4.505000e-09
Reset re-loads circuit *---------------------------------------------
```

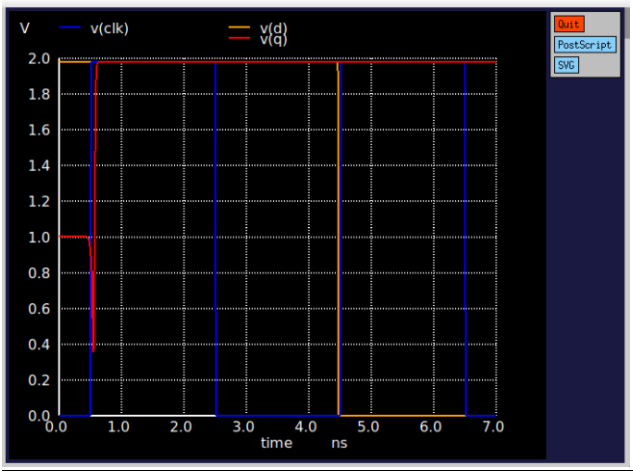
# 4. ff corner & 1.8V

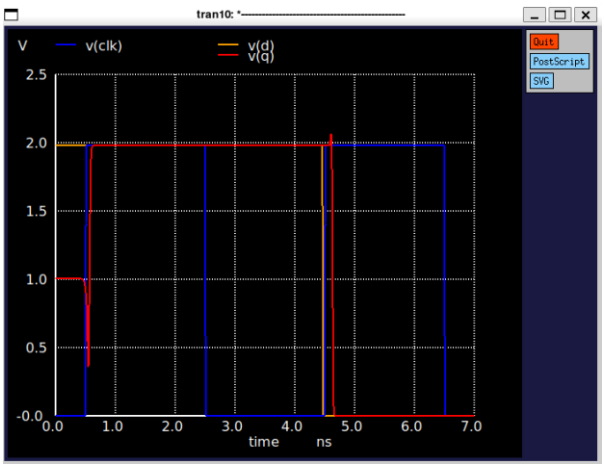



```
 Reference value :  6.99595e-09
No. of Data Rows : 70038
tdc_delay = 4.200000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.000000e-01 rise=2 targ v(q) val=9.000000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                        1.8
xdff.q_n                             1.03472e-07
d                                          1.8
q                                          1.8
clk                                          0
y                                    1.03472e-07
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -8.56922e-10

 Reference value :  6.96825e-09
No. of Data Rows : 70038
tdc_delay = 4.400000e-11
tpcq                =  1.533912e-10 targ=  4.658391e-09 trig=  4.505000e-09
Reset re-loads circuit *---------------------------------------------
```

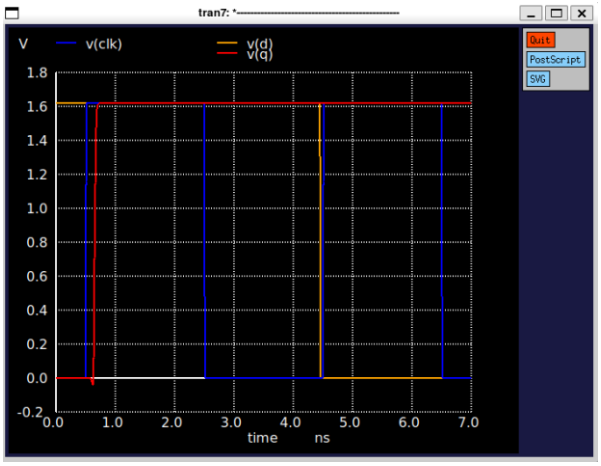
5. ff corner & 1.98V

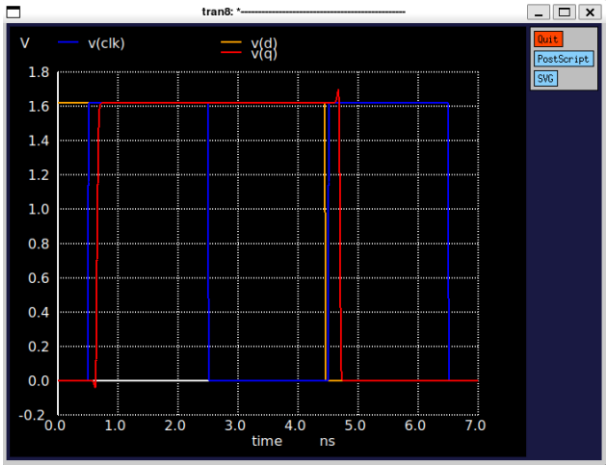



```
 Reference value :  6.98645e-09
No. of Data Rows : 70038
tdc_delay = 3.600000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.900000e-01 rise=2 targ v(q) val=9.900000e-01 fall=2 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.98
xdff.q_n                            0.00238399
d                                         1.98
q                                      1.00589
clk                                          0
y                                    0.0724704
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                        -0.000170144

 Reference value :  6.97445e-09
No. of Data Rows : 70038
tdc_delay = 3.800000e-11
tpcq                =  1.282151e-10 targ=  4.633215e-09 trig=  4.505000e-09
```

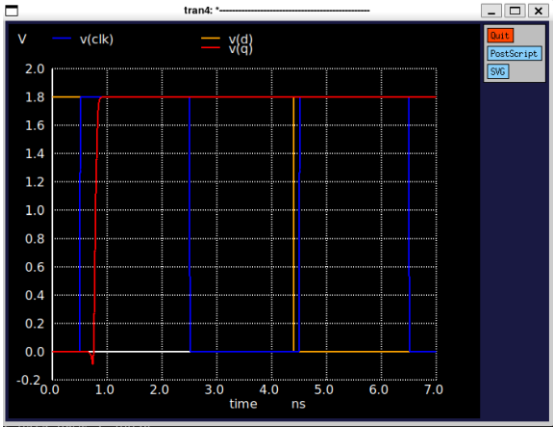
6. ff corner & 1.62V

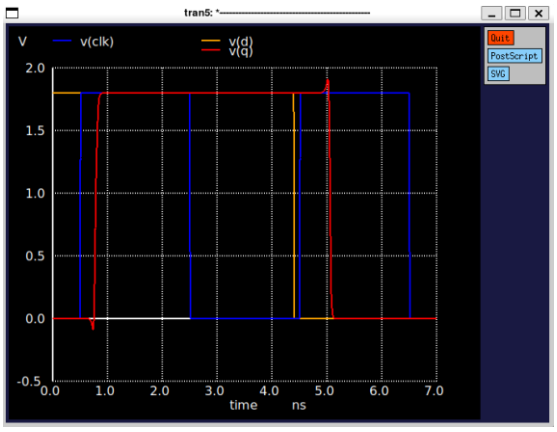


```
 Reference value :  6.99655e-09
No. of Data Rows : 70038
tdc_delay = 5.200000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=8.100000e-01 rise=2 targ v(q) val=8.100000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.62
xdff.q_n                                  1.62
d                                         1.62
q                                    8.73528e-08
clk                                          0
y                                         1.62
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -9.42982e-10

 Reference value :  6.96255e-09
No. of Data Rows : 70038
tdc_delay = 5.400000e-11
tpcq                 =  1.953485e-10 targ=  4.700349e-09 trig=  4.505000e-09
```

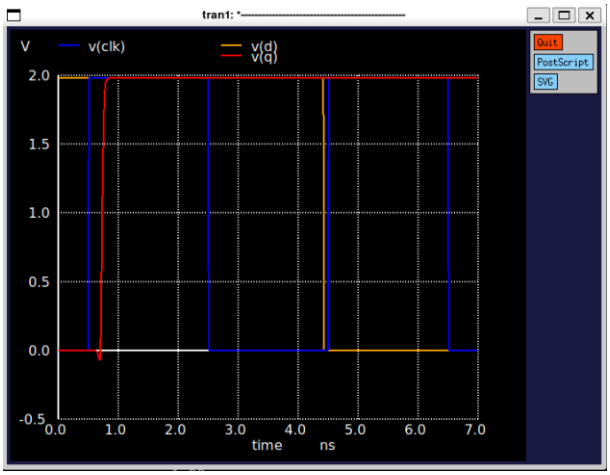
7. ss corner & 1.8V

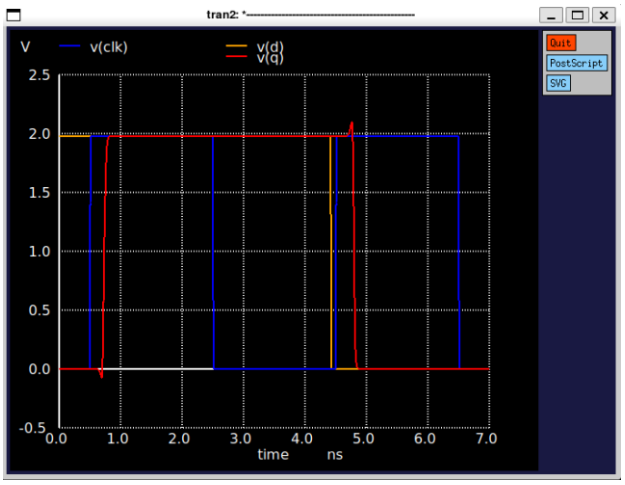



```
 Reference value :  6.99475e-09
No. of Data Rows : 70038
tdc_delay = 1.060000e-10

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.000000e-01 rise=2 targ v(q) val=9.000000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                        1.8
xdff.q_n                                   1.8
d                                          1.8
q                                    4.4618e-09
clk                                          0
y                                          1.8
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -3.7081e-11

 Reference value :  6.99005e-09
No. of Data Rows : 70038
tdc_delay = 1.080000e-10
tpcq                 =  5.538705e-10 targ=  5.058870e-09 trig=  4.505000e-09
```

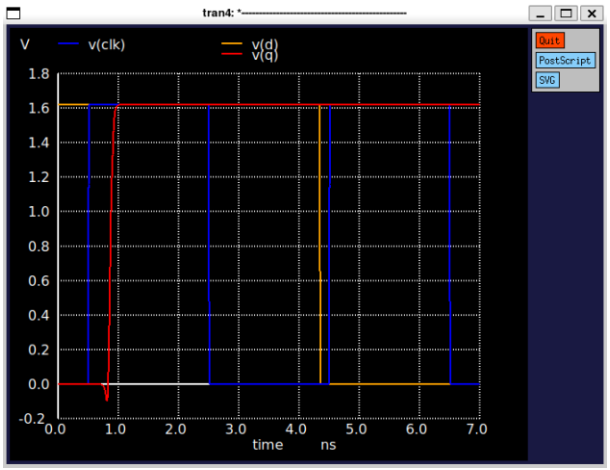
8. ss corner & 1.98V

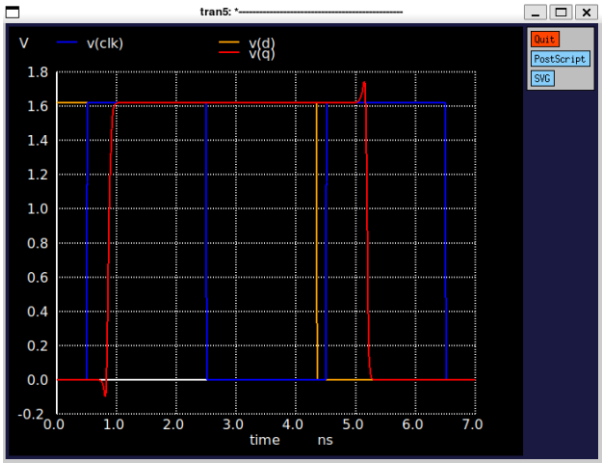



```
 Reference value :  6.95795e-09
No. of Data Rows : 70038
Warning: Missing charsets in String to FontSet conversion
tdc_delay = 8.000000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.900000e-01 rise=2 targ v(q) val=9.900000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.98
xdff.q_n                                  1.98
d                                         1.98
q                                    6.17581e-09
clk                                          0
y                                         1.98
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                           -5.4447e-11

 Reference value :  6.97735e-09
No. of Data Rows : 70038
tdc_delay = 8.200000e-11
tpcq                 =  2.951876e-10 targ=  4.800188e-09 trig=  4.505000e-09
Reset re-loads circuit *---------------------------------------------
```

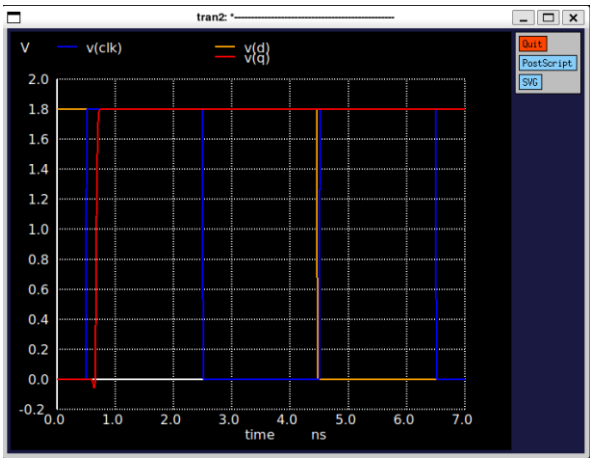
# 9. ss corner & 1.62V

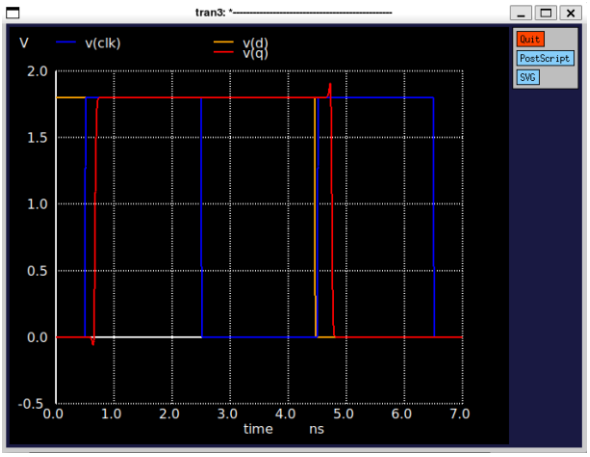


```
 Reference value :  6.99045e-09
No. of Data Rows : 70038
tdc_delay = 1.560000e-10

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=8.100000e-01 rise=2 targ v(q) val=8.100000e-01 fall=1 failed!

Reset re-loads circuit *-----------------------------------------------

Circuit: *-----------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.62
xdff.q_n                                  1.62
d                                         1.62
q                                   3.91335e-09
clk                                           0
y                                         1.62
vdin#branch                                   0
vclk#branch                                   0
vdd#branch                          -2.89162e-11

 Reference value :  6.96675e-09
No. of Data Rows : 70038
tdc_delay = 1.580000e-10
tpcq                 =  6.895360e-10 targ=  5.194536e-09 trig=  4.505000e-09
Reset re-loads circuit *-----------------------------------------------
```

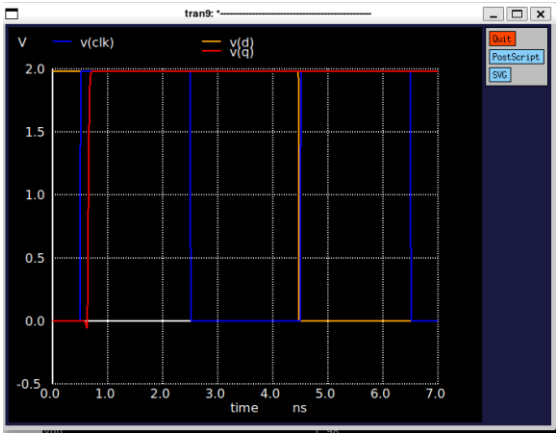
# 10. fs corner & 1.8V

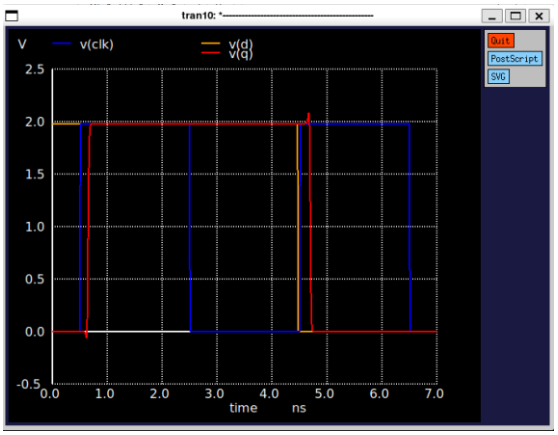



```
 Reference value :  6.99925e-09
No. of Data Rows : 70038
tdc_delay = 4.200000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.000000e-01 rise=2 targ v(q) val=9.000000e-01 fall=1 failed!

Reset re-loads circuit *-----------------------------------------------

Circuit: *-----------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                        1.8
xdff.q_n                                   1.8
d                                          1.8
q                                    1.2566e-06
clk                                          0
y                                          1.8
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -5.41197e-09

 Reference value :  6.98205e-09
No. of Data Rows : 70038
tdc_delay = 4.400000e-11
tpcq                 =  2.481548e-10 targ=  4.753155e-09 trig=  4.505000e-09
Reset re-loads circuit *-----------------------------------------------
```

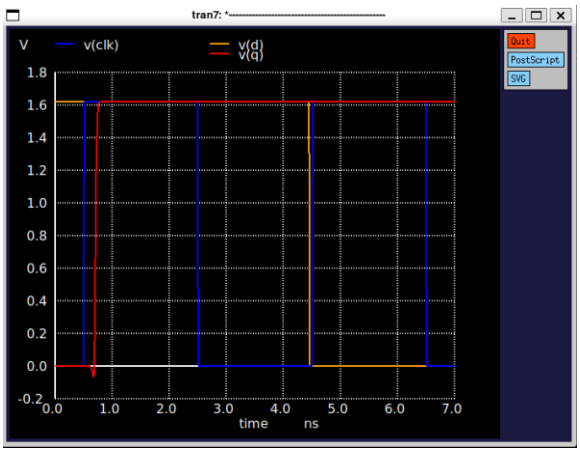
## 11. fs corner & 1.98V

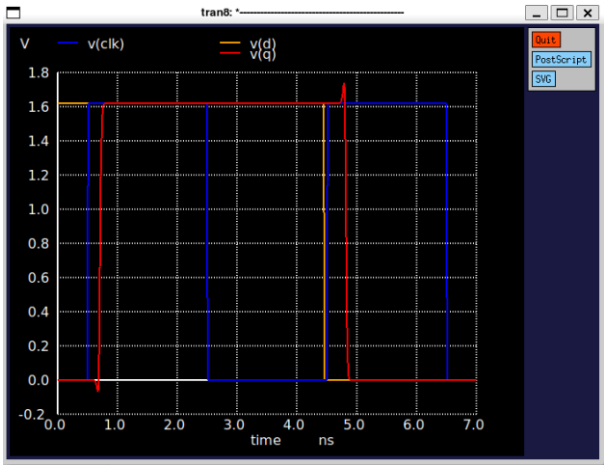



```
 Reference value :  6.95055e-09
No. of Data Rows : 70038
tdc_delay = 3.600000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.900000e-01 rise=2 targ v(q) val=9.900000e-01 fall=1 failed!

Reset re-loads circuit *-----------------------------------------------

Circuit: *-----------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.98
xdff.q_n                                  1.98
d                                         1.98
q                                      1.443e-06
clk                                           0
y                                         1.98
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                            -6.69031e-09

 Reference value :  6.96525e-09
No. of Data Rows : 70038
tdc_delay = 3.800000e-11
tpcq                 =  1.916552e-10 targ=  4.696655e-09 trig=  4.505000e-09
```

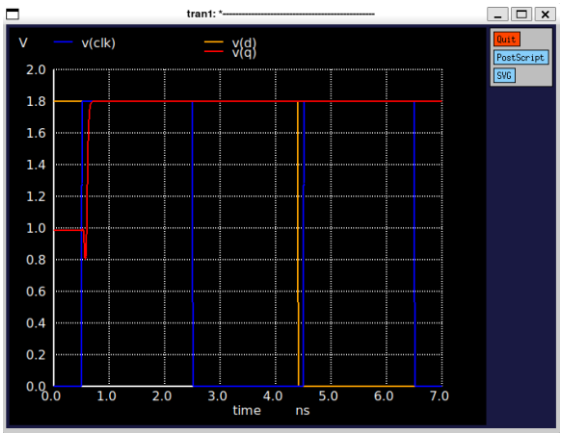
## 12. fs corner & 1.62V

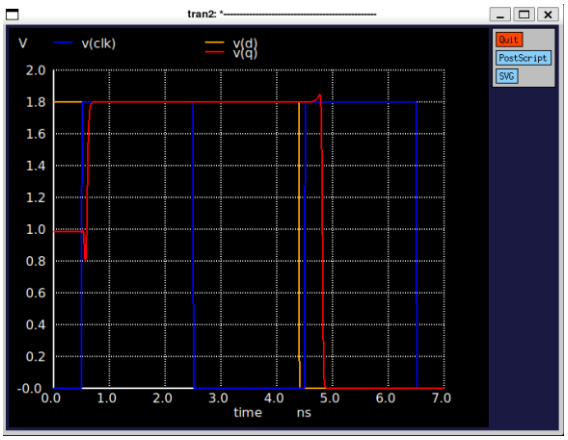


```
 Reference value :  6.97075e-09
No. of Data Rows : 70038
tdc_delay = 5.200000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=8.100000e-01 rise=2 targ v(q) val=8.100000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.62
xdff.q_n                                  1.62
d                                         1.62
q                                   1.13382e-06
clk                                          0
y                                         1.62
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -4.37489e-09

 Reference value :  6.99315e-09
No. of Data Rows : 70038
tdc_delay = 5.400000e-11
tpcq                =  3.169657e-10 targ=  4.821966e-09 trig=  4.505000e-09
Reset re-loads circuit *---------------------------------------------
```

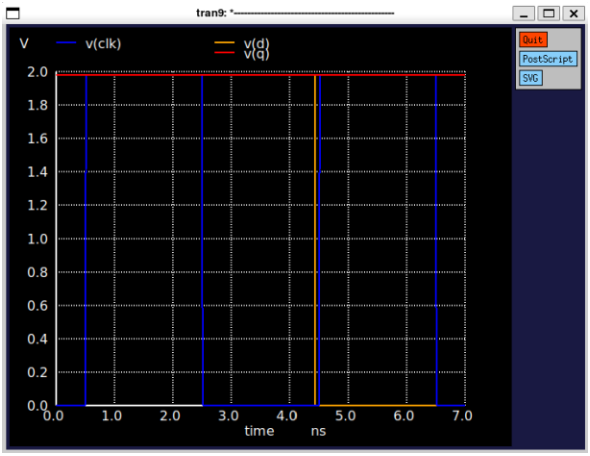
# 13. sf corner & 1.8V

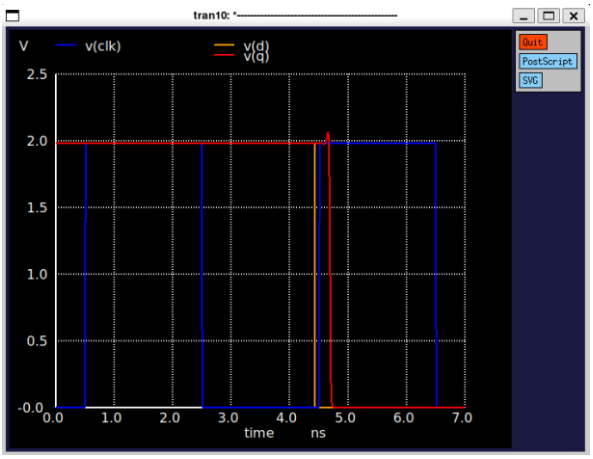



```
 Reference value :  6.97125e-09
No. of Data Rows : 70038
Warning: Missing charsets in String to FontSet conversion
tdc_delay = 1.000000e-10

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.000000e-01 rise=2 targ v(q) val=9.000000e-01 fall=2 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                        1.8
xdff.q_n                             8.8878e-08
d                                          1.8
q                                     0.983653
clk                                          0
y                                    0.00122132
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -3.59478e-05

 Reference value :  6.95355e-09
No. of Data Rows : 70038
tdc_delay = 1.020000e-10
tpcq                =  3.177367e-10 targ=  4.822737e-09 trig=  4.505000e-09
```

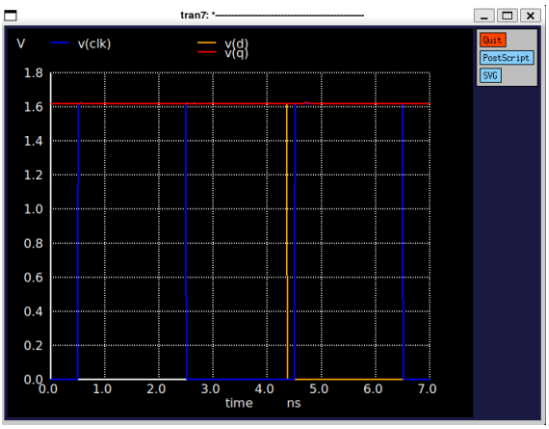
## 14. sf corner & 1.98V

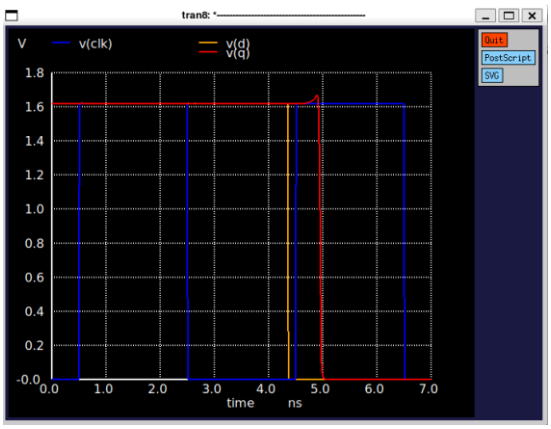



```
 Reference value :  6.96105e-09
No. of Data Rows : 70038
tdc_delay = 7.600000e-11

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=9.900000e-01 rise=2 targ v(q) val=9.900000e-01 fall=1 failed!

Reset re-loads circuit *-----------------------------------------------

Circuit: *-----------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.98
xdff.q_n                             2.66025e-09
d                                         1.98
q                                         1.98
clk                                          0
y                                    2.66025e-09
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -2.51597e-10

 Reference value :  6.94525e-09
No. of Data Rows : 70038
tdc_delay = 7.800000e-11
tpcq                 =  1.878484e-10 targ=  4.692848e-09 trig=  4.505000e-09
```

## 15. sf corner & 1.62V



```
 Reference value :  6.98145e-09
No. of Data Rows : 70038
tdc_delay = 1.420000e-10

Error: measure  tpcq  trig(TARG) : out of interval
 meas tran tpcq trig v(clk) val=8.100000e-01 rise=2 targ v(q) val=8.100000e-01 fall=1 failed!

Reset re-loads circuit *---------------------------------------------

Circuit: *---------------------------------------------

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 70.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                       1.62
xdff.q_n                             1.57501e-09
d                                         1.62
q                                         1.62
clk                                          0
y                                    1.57501e-09
vdin#branch                                  0
vclk#branch                                  0
vdd#branch                          -1.74867e-10

 Reference value :  6.96235e-09
No. of Data Rows : 70038
tdc_delay = 1.440000e-10
tpcq                 =  4.505448e-10 targ=  4.955545e-09 trig=  4.505000e-09
Reset re-loads circuit *---------------------------------------------
```

# Questions:

- In your observation, what is the **sky130_fd_sc_hd__dfxbp_1** setup time?

Answer:

The observed setup time for sky130_fd_sc_hd__dfxbp_1 ranges from 38 ps to 158 ps, depending on the process corner and supply voltage.

- What is the impact of process corners and VDD variations on the DFF's timing and setup time?

Answer:

Both slower process corners and lower supply voltages degrade the setup time of the DFF.

Under the worst-case (SS, 1.62V), setup time increases significantly, while in best-case (FF or FS, 1.98V), it minimizes. Therefore, proper timing margin must be considered in design to ensure robustness across PVT (Process, Voltage, Temperature) variations.